LAB 5:

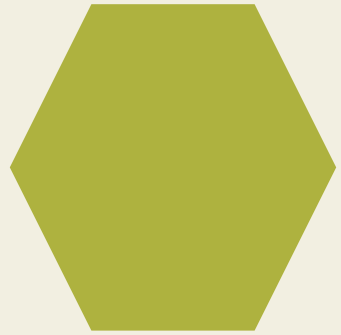
Problem Framing & Modelling II

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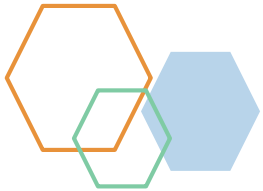
Introduction to Engineering

Fall 2026



TODAY'S LAB · 80 MINUTES

Lab Agenda





Recap

What a model is, the four model types, model anatomy, and assumptions — from Lecture 5.

15 min



Scenario Cards Reused

Same ten scenario cards from Lab 4 / Homework — your team keeps its own card.

5 min



Group Work — Build & Check a Model

Fill the model template: anatomy, Need-Know-How-Solve, assumptions, sanity checks, fit-for-purpose.

45 min · evaluated



Model Answer Walkthrough

Worked model for the EV Charging & Battery Test Station scenario.

15 min

QUICK RECAP · LECTURE 5

What Is a Model?

DEFINITION

A **model** is a simplified representation of a real system, built to answer a specific question.

Four types of models

PHYSICAL

A real copy

A smaller or simpler real object — a turbine blade tested in a wind tunnel.

GRAPHICAL

A diagram

A drawing of parts and links — a single-line diagram or process-flow sketch.

MATHEMATICAL

Equations

Relations between variables — Ohm's law, an energy-balance equation.

COMPUTATIONAL

A calculation

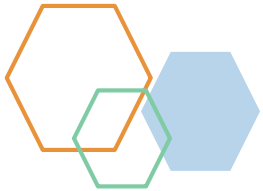
Rules stepped through on a computer — a spreadsheet of hourly demand.

Real projects mix several types: a diagram to organise, equations to calculate, a spreadsheet to run the numbers.

A model is a machine: *numbers go in, a rule is applied, numbers come out. Every number needs its units.*

QUICK RECAP · LECTURE 5

Anatomy of a Model



INPUTS

What we feed in

Values we choose or can change. e.g. battery size, speed, route.

PARAMETERS

What we hold fixed

Numbers describing the system/setting. e.g. energy use per km.

RELATIONSHIPS

The rules

Equations or steps linking the numbers. e.g. Energy = rate × distance.

OUTPUTS

What it predicts

The results we judge. e.g. range (km), energy per day (kWh).

Need–Know–How–Solve

NEED

The variable we want

e.g. range of one charge (km).

KNOW

What we already know

e.g. battery capacity, energy per km, route length.

HOW

Model and assumptions

e.g. Range = usable energy ÷ energy per km.

SOLVE

Do the calculation

Compute, check units, interpret the result.

The How also includes the assumptions you make to solve the problem. Today we build the How.

QUICK RECAP · LECTURE 5

Assumptions & Fit for Purpose

Three kinds of assumptions

SIMPLIFYING

What we ignore

Hills, traffic lights, stops; differences between drivers.

DATA

What we fix

e.g. energy use per km = 1.2 kWh/km (fixed value).

BOUNDARY

What we leave outside

Passengers, charger, the grid — only the system itself is modelled.

COURSE TEMPLATE

We assume **[assumption]** because **[reason]**. If it is wrong, **[effect on the result]**.

Is the model good enough?

TOO SIMPLE

Misses what matters

Key effects (winter, hills, ageing) are invisible to the model.

FIT FOR PURPOSE

Answers the question

Enough detail to decide — no more, no less.

TOO DETAILED

Cannot be built or checked

Needs data or effort the team does not have.

Choose the simplest model that answers the question well enough to decide. Check it with units, order of magnitude, and a known value.

GROUP WORK · EVALUATED · 45 MINUTES



Build & Check a Model

Same scenario card as Lab 4 — now turn it into a working model with stated assumptions.

GROUP WORK · EVALUATED

Your Task: Ten Steps, One Model

Use the same scenario card your team worked with in Lab 4 / Homework. Build and check a model for it.

1. Define one clear model question/purpose
2. Choose a model type and justify it
3. Build the anatomy: Inputs, Parameters, Relationships, Outputs — all with units
4. Complete Need – Know – How – Solve
5. List ≥6 assumptions: simplifying, data, boundary — use the template
6. Mark each assumption Safe / Risky / Need Data
7. Change one assumption; explain the effect on the output
8. Check the model: units, order of magnitude, a known value
9. Judge the model: Too Simple / Fit for Purpose / Too Detailed
10. State one revision and one additional data need

GRADING · PER STEP

3 Good / complete

2 Partial

1 Present, almost nothing done

0 Absent

Graded live across all ten steps. Work through them in order — each later step depends on what you wrote before it.

TEMPLATE · STEPS 1–3

Model Question, Type & Anatomy

1 · MODEL QUESTION

What one question should this model answer? Write it as a question.

2 · MODEL TYPE

Physical / Graphical / Mathematical / Computational — and why it fits.

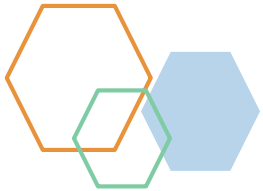
3 · Build the anatomy — every entry needs a unit

Part	Name it (with unit)	Role
Inputs	…	What we feed in / can choose
Parameters	…	What we hold fixed
Relationships	…	The equation(s) or rule(s)
Outputs	…	What the model predicts

A model is a machine: numbers go in, a rule is applied, numbers come out. Every number needs its units.

TEMPLATE · STEPS 4–6

Need–Know–How–Solve & Assumptions



4 · Complete Need – Know – How – Solve

NEED	KNOW	HOW	SOLVE
…	…	…	…

5 · List at least 6 assumptions

TEMPLATE — WRITE 6 OF THESE

We assume **[assumption]** because **[reason]**. If it is wrong, **[effect on the result]**.

6 · Classify each assumption

Assumption (short)	Kind (simplifying/data/boundary)	Safe / Risky / Need Data
…	…	…
…	…	…

Safe = holds under normal conditions. Risky = could fail and change the answer. Need Data = you cannot tell yet.

TEMPLATE · STEPS 7–10

Stress-Test & Judge Your Model

7 · Change one assumption — what happens to the output?

Assumption changed	Old value	New value	Effect on output
…	…	…	…

8 · Sanity-check the model

UNITS

Do the units of the result work out correctly?

ORDER OF MAGNITUDE

Is the number plausible — tens, not millions?

KNOWN VALUE

Does it roughly match a datasheet or known case?

9 · Judge the model

- Too Simple
- Fit for Purpose
- Too Detailed

10 · One revision and one data need

REVISION

One concrete change you would make to the model.

ADDITIONAL DATA NEED

One piece of real data that would make it more trustworthy.



EV Charging & Battery Test Station

A worked model applying all ten steps of the template.

1 EV Charging and Battery Test Station



An EV charging and battery test station receives electric vehicles for fast charging and battery health checks. Vehicles arrive at an average rate of 12 cars per hour and wait in a queue if all 4 chargers are busy. Each charge takes about 30–40 minutes, followed by a 10-minute battery diagnostic test. If the battery health is below 80%, a technician inspects it. Customers want shorter waiting times, operators are concerned about electricity demand peaks, and the local grid operator worries about high power draw. Some drivers think the station is slower than competitors, possibly due to conservative safety limits.

Question, Type & Anatomy

1 · MODEL QUESTION

Will the station's peak power draw stay within the grid operator's 150 kW limit during the busiest hour?

2 · MODEL TYPE — MATHEMATICAL

Chosen because the question is a direct calculation: power per charger × chargers in use, compared with a limit.

3 · Model anatomy

Part	Name (with unit)	Role
Inputs	Chargers in use n [count]; power per charger P_1 [kW]	What we feed in / can choose
Parameters	Grid power limit L = 150 [kW]; number of chargers = 4	What we hold fixed
Relationships	Total draw: $P = n \times P_1$; compliant if $P \leq L$	The equation / rule
Outputs	Total power draw P [kW]; compliance (yes/no)	What the model predicts

4 · Need – Know – How – Solve

NEED

Peak power draw, in kW

KNOW

P_1 = 22 kW/charger; 4 chargers; limit 150 kW

HOW

$P = n \times P_1$, with n = 4 at peak

SOLVE

$P = 4 \times 22 = 88$ kW ≤ 150 kW ✓

Assumptions & One Changed Case

We assume … because …	Kind	Risk	If it is wrong …
All 4 chargers draw 22 kW simultaneously at peak, because that is the rated power per unit.	Data	Safe	Actual peak draw would be lower, not higher.
Charging power is constant through the session, because we ignore the taper near full charge.	Simplifying	Risky	Real peak draw is slightly lower than modelled.
The diagnostic test draws negligible power, because it is a low-current measurement.	Simplifying	Safe	Model would slightly underestimate draw.
Only the 4 charger bays are in the model's boundary, because facility lighting/HVAC is metered separately.	Boundary	Need Data	Facility load could add to the grid connection total.
Battery health/inspection does not add electrical load, because it uses the charger's own diagnostics.	Boundary	Safe	No change expected to the result.
All 4 chargers can be busy at once at peak hour, because that is the worst case, not the average.	Data	Risky	If fewer than 4 are ever busy together, draw is lower.

One changed assumption — what if it's wrong?

Changed: chargers are upgraded to fast-charge mode, so P_1 rises from 22 kW to 50 kW per charger.

New result: $P = 4 \times 50 = 200$ kW — now above the 150 kW limit. The model flags a compliance failure the original assumption hid.

Sanity Checks & Final Judgement

Sanity checks

Check	Result
Units	n [count] × P_1 [kW] = P [kW] — units are consistent.
Order of magnitude	88 kW for 4 fast chargers is plausible; not 8 kW (too low) or 8,000 kW (too high).
Known value	Matches a public DC fast-charger datasheet (~20–25 kW per standard unit).

Final fit-for-purpose judgement

Too Simple | **Fit for Purpose** | Too Detailed

Chosen: Fit for Purpose — the model answers the compliance question (yes/no against 150 kW) with the data on hand, without needing a sub-minute load-curve simulation.

REVISION

Replace the "all 4 busy at once" worst case with a measured diversity factor from real usage logs.

ADDITIONAL DATA NEED

A week of metered power-draw data per charger, to confirm the 22 kW rating holds in practice.

THANK YOU

Questions?

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